

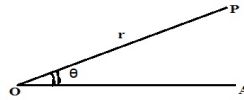
Lecture 18: Curve Tracing

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Polar Co-ordinates

The position of a point P on a plane can also be indicated by stating (i). its distance r from a fixed point O and (ii) the inclination θ of OP to a fixed straight line OA .

r and θ are called polar co-ordinate of P ; r is called the radius vector and θ the



vertical angle, O is pole and OA is the initial line. r is considered to be positive when measured away.

1. **Solve for r :** Solve the equation for r and consider how r varies as θ increases from some value θ_1 to $+\infty$ and also as θ diminishes from θ_1 to $-\infty$.
2. **Symmetry:** If changing the sign of θ does not change the value of r , the curve is symmetrical about the initial line.

Similarly, if changing r to $-r$ and θ to $-\theta$ does not alter the given equation or if changing θ to $\pi - \theta$ does not alter the value of r , the curve is symmetrical about the line $\theta = \pi/2$.

3. In most of the polar equations only periodic functions ($\sin \theta, \cos \theta$ etc.) occur, and so values of θ from 0 to 2π (or sometimes some multiple or submultiple of 2π) need alone be considered. The remaining values of θ give no new branches of the curve.
4. **Direction of tangent:** Form $\tan \phi = r \frac{d\theta}{dr}$, determine ϕ if feasible. This will give the direction of the curve at some representative points.

Example. Trace the curve $r = a \sin 3\theta$.

Solution.

- As $\theta \Leftrightarrow \pi - \theta \Rightarrow r = a \sin(\pi - \theta) = a \sin 3\theta$, so the curve is symmetrical about the line $\theta = \frac{\pi}{2}$.

- $r = 0$ gives $a \sin 3\theta = 0 \Rightarrow 3\theta = n\pi$.

$$\Rightarrow \theta = \frac{n\pi}{3} = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}.$$

So, the curve passes through the pole.

Tangent at the pole :

$$\tan \phi = r \frac{d\theta}{dr} = 0 \Rightarrow r = 0 \text{ as } \frac{d\theta}{dr} \neq 0.$$

$$r = 0 \Rightarrow \sin 3\theta = 0 = 3\theta = n\pi$$

So, $\theta = 0, \frac{\pi}{3}, \frac{2\pi}{3}, \pi, \frac{4\pi}{3}, \frac{5\pi}{3}$, are tangent at the pole.

- Since, maximum $r = a$ so curve lies inside a circle at pole and radius a .
- $\frac{dr}{d\theta} = 3a \cos 3\theta$

$$\frac{dr}{d\theta} > 0 \forall 3\theta \in \left[0, \frac{\pi}{2}\right] \cup \left[\frac{3\pi}{2}, 2\pi\right] \Rightarrow \theta \in \left[0, \frac{\pi}{6}\right] \cup \left[\frac{\pi}{3}, \frac{\pi}{2}\right],$$

and,

$$\frac{dr}{d\theta} < 0 \forall 3\theta \in \left[\frac{\pi}{2}, \pi\right] \cup \left[\pi, \frac{3\pi}{2}\right] \Rightarrow \theta \in \left[\frac{\pi}{6}, \frac{\pi}{3}\right] \cup \left[\frac{\pi}{3}, \frac{\pi}{2}\right]$$

$\Rightarrow r$ increases from 0 to a as θ increases from 0 to $\frac{\pi}{6}$ and r decreases from a to 0 in the interval $\left[\frac{\pi}{6}, \frac{\pi}{2}\right]$. r increases from 0 to a numerically lies in 3^{rd} quadrant as θ increases from $\frac{\pi}{3}$ to $\frac{\pi}{2}$.

Example. Trace the curve $r = a(1 + \cos \theta)$.

Solution. (i). The curve is symmetrical about initial line, since its equation remains unchanged by replacing θ by $-\theta$.

(ii). When $\theta = 0$, $r = 2a$ and when $\theta = \pi$, $r = 0$.

Also, when $\theta = \pi/2$, $r = a$ and when $\theta = \frac{3\pi}{2}$, $r = a$.

Now,

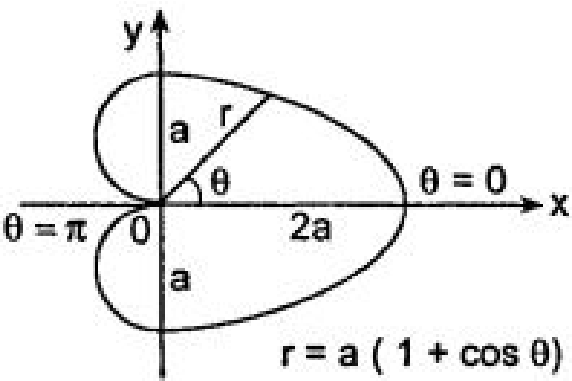
$$\tan \phi = \frac{rd\theta}{dr} = \frac{a(1 + \cos \theta)}{-a \sin \theta}.$$

$$\Rightarrow \tan \phi = -\cot \left[\frac{\theta}{2}\right] = \tan \left[\frac{\pi}{2} + \frac{\theta}{2}\right].$$

$$\psi = \theta + \phi = \theta + \frac{\pi}{2} + \frac{\theta}{2} = \frac{3\theta}{2} + \frac{\pi}{2}.$$

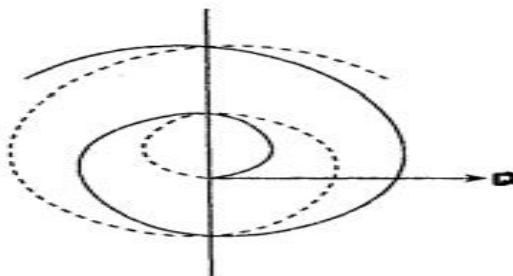
When $\theta = 0$, $\psi = \frac{\pi}{2}$ i.e., the tangent at this point is perpendicular to the initial line. When $\theta = \pi$, $\psi = 2\pi$, i.e., the tangent at this point coincide with initial line. The following table gives some values of r and θ .

θ	0	$\pi/3$	$\pi/2$	$2\pi/3$	π
r	$2a$	$3a/2$	a	$a/2$	0



Example. Trace the curve $r = a\theta$.

Solution. Let us consider first only positive values of θ . When $\theta = 0, r = 0$, so the curve passes through the pole. As θ increases through $\frac{\pi}{2}, \pi, \frac{3\pi}{2}, 2\pi, \dots$ we see that r increases and the point tracing the curve goes away from the pole as shown in the figure. When $\theta \rightarrow \infty, r \rightarrow \infty$.



If we consider negative values of θ also we get a curve shown by dotted line.

Reference Book: Differential Calculus, Dr. Gorakh Prasad. Pothishala Private LTD.